MAC 233: External LED Lab

Dr Max Champneys

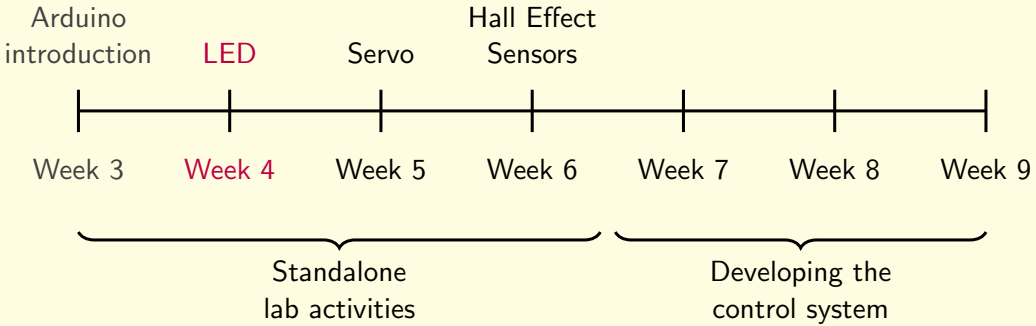
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MAC 233: **21/10/2026**



University of Sheffield

MAC 233: LABS OVERVIEW



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TODAY'S LAB: PREREQUISITES

This worksheet builds upon the Lab 1 worksheet. You will have needed to complete the Lab 1 worksheet before you can continue with this lab.

Out of your Arduino pack, you will need: 1 Arduino, 1 breadboard, 1 LED, 1 220Ω resistor, and 1 jumper cable.

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TODAY'S LAB: OBJECTIVES

The objective of today's lab is to learn how to use an electronic breadboard to create a circuit and to control an external output on the Arduino.

We will be taking the scenario from the previous worksheet and changing it to communicate with an external LED.

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REMINDER: 'PINMODE' AND 'DIGITALWRITE'

`pinMode` sets the IO direction of a pin and belongs in `setup`.

`digitalWrite` then sets the pin HIGH (+5V) or LOW (0V).

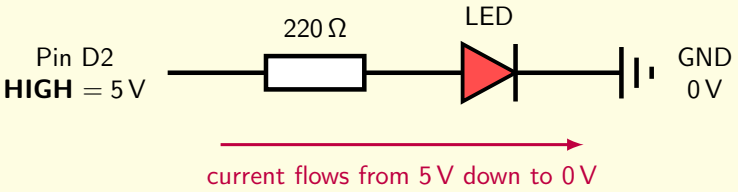
```
#define LED_PIN 2

void setup() {
  pinMode(LED_PIN, OUTPUT);      // once, in setup
}

void loop() {
  digitalWrite(LED_PIN, HIGH);  // 5 V, LED on
  delay(1000);
  digitalWrite(LED_PIN, LOW);   // 0 V, LED off
  delay(1000);
}
```


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WHAT ‘HIGH’ ACTUALLY DOES

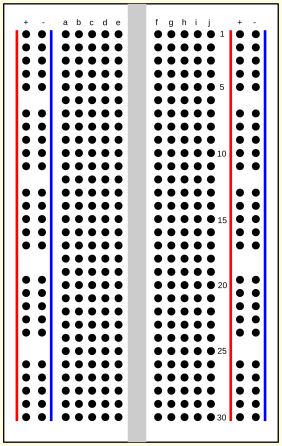


`digitalWrite(LED_PIN, HIGH)` connects the pin to 5 V, and ‘LOW’ connects it to 0 V. It is that 5 V that pushes a current through the LED and lights it.

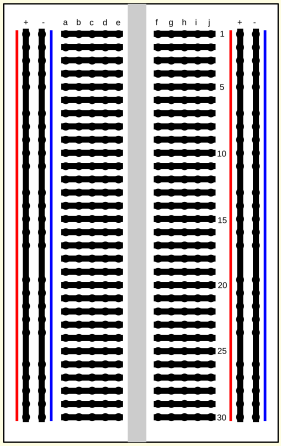
The LED itself uses up about 2 V, so the resistor takes the other 3 V. With no resistor, nothing limits the current and both the LED and the pin can be damaged.

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THE BREADBOARD



The board as you see it.

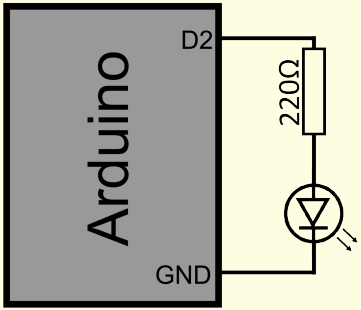


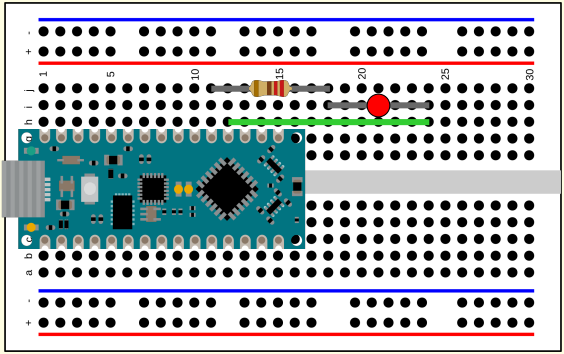
The hidden connections drawn in.

Breadboard source: `https://freesvg.org/breadboard`.

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WIRING DIAGRAM



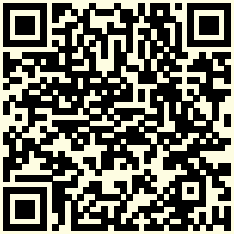


Breadboard source: `https://freesvg.org/breadboard`, Resistor source `https://freesvg.org/resistor-330-ohm`.

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HELP

Lab 2 worksheet



Nano Every pinout



Problems? Raise your hand and someone will come and help you.

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